Survival Analysis on Employee Attrition — Full Interview-Ready Explanation

1. Why Survival Analysis (Not Just Classification)?

Most people approach attrition as a binary classification problem — "will this employee leave, yes or no?" I framed it differently: **"how long until they leave, and what factors speed that up or slow it down?"** This is a fundamentally richer question because it incorporates *time* as a dimension, and it naturally handles employees who haven't left yet — which classification can't do cleanly.

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2. Theoretical Foundation — Survival Analysis

Survival function S(t) — the probability that an employee is still with the company after t years. Formally: $S(t) = P(T > t)$, where T is the random variable representing time until departure. It starts at 1 (everyone present at t=0) and monotonically decreases toward 0.

Hazard function h(t) — the instantaneous rate of leaving at time t, given survival up to t. While S(t) tells you "what fraction remain," h(t) tells you "how dangerous is *this exact moment* for the people still here." Formally:

$$h(t) = \lim(\Delta t \to 0) [P(t \le T < t+\Delta t \mid T \ge t)] / \Delta t$$

The survival and hazard functions are linked: $S(t) = \exp(-\int_0^t h(u)du)$. This means if you know either function completely, you can derive the other — they are two views of the same underlying phenomenon.

Cumulative hazard H(t) — the accumulation of hazard over time: $H(t) = -\log S(t)$. Useful for model diagnostics and for understanding total accumulated risk up to time t.

Censoring — the most important concept separating survival analysis from regular regression. An observation is right-censored when an employee is still employed at the time of data collection — we know they survived *at least* this long, but their eventual departure time is unknown. Dropping these observations or treating them as "never leaving" would massively bias results by ignoring all the information in their observed tenure. Every method used here correctly incorporates censored observations. In this dataset: 237 employees left (events), 1,233 are right-censored (still employed).

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3. The Lomax-Gompertz Distribution — What It Is and Why It Was Created

Background: The Gap in Lifetime Modelling

In survival analysis and reliability engineering, choosing the right parametric distribution to model lifetime data is critical. Classical distributions each have limitations:

Exponential distribution — the simplest survival model. It assumes a *constant* hazard rate $h(t) = \lambda$ for all t — meaning the risk of failure/departure is the same at every moment, regardless of how long the subject has already survived. This is the "memoryless" property. In practice this is almost never realistic — employees who just joined are not equally likely to leave as those who have survived 10 years, and a machine that has just been replaced is not equally likely to fail as one that has been running for years.

Weibull distribution — extends the exponential by allowing the hazard to increase or decrease monotonically over time: $h(t) = (k/\lambda)(t/\lambda)^{(k-1)}$. With $k > 1$ the hazard increases (aging/wear-out), with $k < 1$ it decreases (infant mortality / early learning curve effects). But it cannot capture a hazard that first increases then decreases, or vice versa — it is always monotone.

Gompertz distribution — developed originally to model human mortality. Its hazard function is $h(t) = \alpha e^{(\beta t)}$ — an exponentially increasing hazard, reflecting how mortality risk accelerates with age. Very useful in biological survival data but inflexible for engineering or business contexts where hazard shapes may differ.

Lomax distribution — also called the Pareto Type II distribution. Has a heavy right tail, useful for modelling data with extreme values and for reliability in engineering contexts. Its hazard is $h(t) = \alpha/(1 + \lambda t)$, a decreasing function — useful for infant-mortality type scenarios.

The Problem Each Alone Cannot Solve

Real-world lifetime data — whether patient survival times, machine failure times, or product lifetimes — often exhibit **non-monotone hazard shapes**: hazard initially increases (early vulnerability) then decreases (survivors become more robust), forming a bathtub or hump-shaped curve. Neither Weibull, Gompertz, nor Lomax alone can capture this.

What the Lomax-Gompertz Distribution Does

The **Lomax-Gompertz distribution** is a compound distribution created by combining the Lomax and Gompertz distributions. The motivation is to produce a more flexible family that can model a wider range of hazard shapes — including non-monotone ones — that neither parent distribution can handle alone.

The construction uses a technique called **compounding** or **mixing** — where one distribution's parameter is itself treated as a random variable following another distribution. This generates a new distribution with a richer parametric structure. The resulting hazard function can exhibit increasing, decreasing, or bathtub-shaped behavior depending on the parameter values, making it applicable across a much broader range of real-world scenarios.

Key properties derived and studied in my paper:

- The probability density function (PDF) and cumulative distribution function (CDF)
- The survival function S(t) and hazard function h(t) in closed form
- Statistical moments — mean, variance, skewness, kurtosis
- Parameter estimation via **Maximum Likelihood Estimation (MLE)** — finding the parameter values that make the observed data most probable under the assumed distribution
- **Goodness-of-fit testing** — using criteria like AIC (Akaike Information Criterion) and BIC (Bayesian Information Criterion) to compare the Lomax-Gompertz against alternative distributions on real lifetime datasets, and checking graphically using probability-probability (PP) and quantile-quantile (QQ) plots

Why create a new distribution rather than use existing ones? Because in parametric survival modelling, model misspecification matters. If the true hazard shape is bathtub-shaped and you fit a Weibull (which is monotone), your estimates of survival probabilities, median survival times, and covariate effects will all be wrong. The Lomax-Gompertz provides a more flexible alternative that can reduce this misspecification risk, potentially giving better predictions and more accurate inferences in the right data contexts.

Presented at: Kerala Statistical Association International Conference, Vimala College, Thrissur — connecting it to the broader statistical community working on distribution theory and lifetime data analysis.

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4. How This Connects to the Current Project

This is the key bridge that makes the combination of my conference paper and this project compelling as a portfolio:

Dimension	Lomax-Gompertz Paper	IBM Attrition Project
Approach	Fully parametric — assumes a specific distributional form for h(t)	Semi-parametric (Cox) and non-parametric (KM) — makes minimal assumptions
Hazard function	Derived in closed form from the compound distribution	Baseline hazard left completely unspecified in Cox model
Estimation	Maximum Likelihood Estimation	Partial likelihood (Cox), product-limit estimator (KM)

Dimension	Lomax-Gompertz Paper	IBM Attrition Project
Strength	More statistically efficient when the assumed form is correct	More robust when the true distribution is unknown or complex
Weakness	Misspecification risk if real data doesn't match the assumed form	Cannot extrapolate beyond observed data range
Application	Theoretical distribution fitting on lifetime datasets	Applied analysis on real business HR data with covariates

The deliberate point I make at interviews: **I chose Cox over a parametric model for the IBM data because the HR dataset has a complex covariate structure — overtime, satisfaction, income all interact — and imposing a parametric hazard form adds unnecessary assumptions on top of an already complex modelling problem. The semi-parametric Cox approach is the statistically principled choice here.** This shows I don't just apply methods mechanically — I understand when each is appropriate.

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5. Method 1 — Kaplan-Meier Estimator

What it is: A non-parametric estimator of the survival function — no distributional assumptions whatsoever. It is a step function that drops only at observed event times.

Formula: $\hat{S}(t) = \prod(t_i \leq t) [(n_i - d_i) / n_i]$

where n_i = employees at risk just before time t_i, d_i = departures at time t_i. At each event time, survival is multiplied by the fraction of at-risk employees who survived that moment.

Confidence intervals are computed using Greenwood's formula, which estimates the variance of the log survival estimate and produces the shaded bands visible in the plots — wider bands at later times reflect smaller risk sets and greater uncertainty.

What I found:

Overall curve: Survival declines gradually — ~87% at year 5, ~78% at year 10. Median survival reported as 40 years because with only 16.1% overall attrition the curve never crosses 0.5 — this is a censoring artifact, not a literal statement that employees stay 40 years.

Overtime stratification: Curves diverge from year 1. By year 10: ~55% survival (overtime) vs ~80% (no overtime) — a 25 percentage point gap that persists indefinitely. Log-rank $p < 0.0001$.

Department stratification: Sales shows weakest retention (~50% by year 20), R&D strongest, HR in between. However, this proved to be confounded once Cox regression controlled for individual-level factors.

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6. Method 2 — Log-Rank Test

What it is: A non-parametric hypothesis test comparing the full survival distributions of two or more groups — not just a single time point, but the entire curve.

Hypotheses: H_0: $S_1(t) = S_2(t)$ for all t (identical survival distributions). H_1: the groups have different survival experiences.

Mechanics: At each event time t_i, the test computes how many events were *observed* in each group vs how many would be *expected* if group membership had no effect (i.e., if H_0 were true). These observed-minus-expected values are accumulated across all event times and combined into a chi-squared test statistic. The log-rank test gives equal weight to all time points — if you wanted to emphasise early or late differences, weighted variants (Breslow, Tarone-Ware) exist.

Result: $p < 0.0001$ for overtime vs no-overtime — essentially certain that the survival distributions differ.

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7. Method 3 — Cox Proportional Hazards Regression

The model: $h(t|X) = h_0(t) \cdot \exp(\beta_1X_1 + \beta_2X_2 + \ldots + \beta_pX_p)$

- **$h_0(t)$** is the baseline hazard — completely unspecified, estimated from data non-parametrically. This is what makes Cox "semi-parametric."
- The exponential term is each individual's **personal multiplier** on that baseline risk based on their specific characteristics.

Why the exponential form? It ensures the hazard is always positive (you can't have a negative instantaneous risk), and it makes the model multiplicative — covariates act as scaling factors on the baseline.

Estimation — Partial Likelihood: Cox's key insight was that you don't need to estimate $h_0(t)$ to estimate the β coefficients. He showed you can write a "partial likelihood" — a function of only the β's, based on the *ordering* of event times rather than their exact values — and maximise that. This is why Cox regression is computationally tractable without specifying a distributional form.

The Proportional Hazards Assumption: The hazard *ratio* between any two individuals is constant over time. If overtime triples one employee's risk at year 1, it triples it at year

10 too. This can be checked using Schoenfeld residuals — plotting them against time should show no systematic trend if the assumption holds.

Concordance (C-statistic): Analogous to AUC in classification. For any two randomly chosen employees where one experienced attrition before the other, concordance = probability the model correctly assigns higher predicted hazard to the one who left first. **C = 0.82 here — excellent predictive discrimination.**

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8. Full Results

237 events, 1,470 observations, concordance = 0.82, overall model p < 0.001

Variable	HR	95% CI	p-value	Plain English
OverTime	**3.24**	2.50–4.19	<0.005	Overtime employees face **3.24× the attrition hazard** at any moment
JobSatisfaction	0.77	0.68–0.86	<0.005	Each 1-point improvement → 23% lower hazard
WorkLifeBalance	0.78	0.65–0.94	0.01	Each 1-point improvement → 22% lower hazard
Age	0.95	0.93–0.97	<0.005	Each additional year → 5% lower hazard
MonthlyIncome	~1.00*	1.00–1.00	<0.005	Significant protective effect; z = −7.92 confirms strong significance despite near-zero coefficient due to raw currency scale
Department_R&D	0.70	0.38–1.26	0.23	Not significant — CI crosses 1
Department_Sales	1.36	0.74–2.51	0.32	Not significant — CI crosses 1

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9. The Confounding Insight

The KM department curves *suggested* Sales has worse retention and R&D has better. But Cox regression, controlling simultaneously for overtime, satisfaction, income, and age, found **department is not an independent predictor** (both p > 0.2). This means Sales employees appear to leave faster not because of something intrinsic to being in

Sales, but because Sales roles happen to involve higher overtime rates and lower satisfaction — and *those individual conditions* drive the attrition. This is the distinction between a **marginal effect** (what KM shows — unadjusted, everything pooled) and a **conditional effect** (what Cox shows — each variable's effect holding everything else fixed). Recognising this shows genuine statistical maturity.

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10. Business Interpretation

Overtime (HR = 3.24) dwarfs every other variable by a factor of 4 or more. The practical recommendation is unambiguous: reducing mandatory overtime is the single highest-leverage retention intervention available, independent of department, age, or compensation. Job satisfaction and work-life balance are real but secondary levers (~22% effect each). Department-targeted programs are likely less impactful than they appear in raw data, since underlying working conditions — not the department label — are the true drivers.

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11. How I Would Summarise This at an Interview

"My conference paper worked on the theoretical end of survival analysis — constructing a new parametric distribution, the Lomax-Gompertz, by compounding two existing ones to produce a more flexible hazard shape, then studying its properties and fitting it to lifetime data using MLE. That gave me deep fluency with the mathematical structure of survival models — what survival and hazard functions actually represent, and why distributional choices matter.

This project deliberately sat at the applied end of the same spectrum. I used non-parametric and semi-parametric methods — Kaplan-Meier and Cox regression — precisely because the IBM dataset has a complex covariate structure where imposing a parametric hazard form would add unnecessary assumptions. Together, the two pieces show I can work across the full range: from deriving theoretical distributions to applying robust, assumption-light models to real business data and translating statistical findings into actionable recommendations.

The headline finding was an overtime hazard ratio of 3.24 with concordance of 0.82 — a strong, well-fitted model with a clear, interpretable business implication. And the department confounding story shows the analysis went beyond surface correlations to identify true independent drivers of attrition."